housing and the tip of gear. In gear pumps two types of gears are generally used, they are 1) Spur gear and 2) Helical gear. To reduce noise in pump, helical gear is preferred.

During functioning of engine, the drive gear gains the power from the skew gear of the camshaft via distributor extension shaft. The driven gear attached to it also rotates due to the rotation of the drive gear as it is in mesh with the drive gear. However the rotation is opposite. Because of opposite rotation of the two gears, inner side of the pump creates vacuum. Due to this vacuum the oil in the sump tries to fill the empty space. The oil sucked into the housing of the pump through the inlet. In this way, the oil passes through the gears and housing gaps to fill the empty space via inlet. The oil pressure is increased here to about 2 kg/cm^2 to 4 kg/cm^2 and it comes out through the outlet. Since outlet is connected to the oil gallery, the oil is circled through all the moving parts of engine. Figure 8.5.2(a) shows the view of the gear type oil pump.

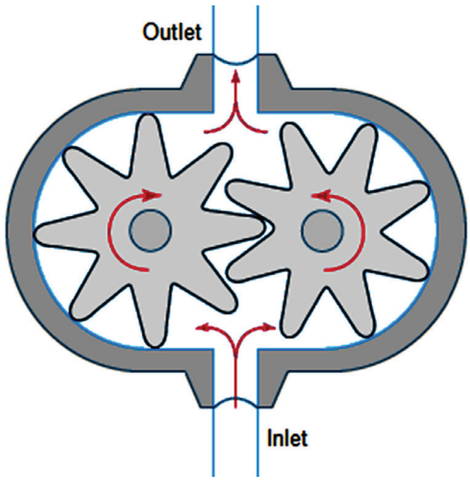


Figure 8.5.2(a) Gear pump

8.5.2.2 Rotor Pump

Rotor pump is similar to the gear type pump. But in this pump two rotors are present instead of gears. These two rotors are attached to the inner side. One of the rotors is called inner rotor and the other called as outer rotor. The inner rotor rotates inside the outer rotor. The outer rotor has one lobe more than that of the inner rotor. The rotating arrangement of the rotors in this pump varies. So the size of the gap in between the two rotors varies. Inner shaft is rotated by attaching to the skew shaft of the cam shaft through the distributor extension shaft. Hence both the rotors rotate. Because of opposite rotation of the two rotors, inner side of the pump creates vacuum. Due to the vacuum pressure, the oil in the sump is forced to enter into the empty space. The oil attains high pressure of about 2 kg/cm^2 to 5 kg/cm^2 at the pump and comes out via the outlet port. Figure 8.5.2(b) shows the view of a rotor pump

1. This pump has 25% more power than gear pump and the construction is very simple.
2. For every rotation as less number of lobs is meshing it is noise free operation.

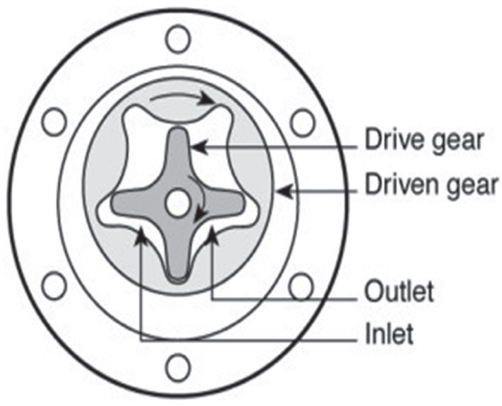


Figure 8.5.2(b) Rotor pump

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